

Problem 1. Consider the line in $\mathbb{R}^3$ passing through the two points

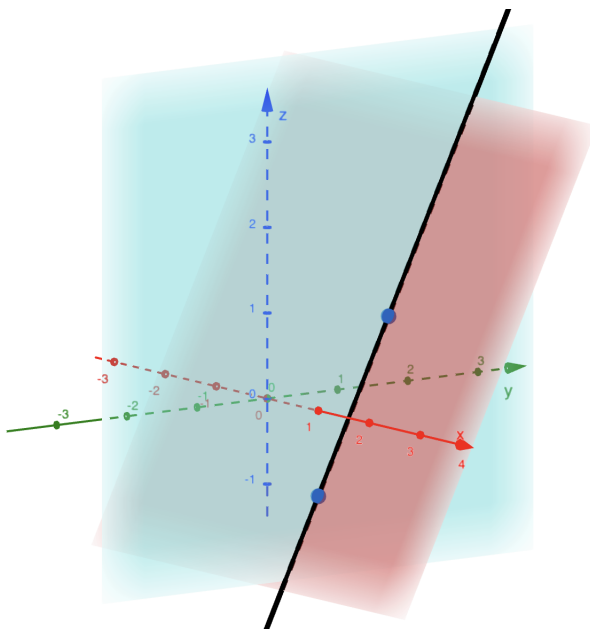
$$P = (1, 0, -1) \quad \text{and} \quad Q = (1, 1, 1).$$

Express this line as an intersection of two planes.

Solution: There are infinitely many planes containing this line. Any two of them will do. The most systematic way is to first consider a parametrization, for example

$$\begin{aligned} \mathbf{r}(t) &= P + t\vec{PQ} \\ &= (1, 0, -1) + t[(1, 1, 1) - (1, 0, -1)] \\ &= (1, 0, -1) + t(0, 1, 2) \\ &= (1, t, 2t - 1). \end{aligned}$$

Then we eliminate the parameter. Since $x(t) = 1$, $y(t) = t$, $z(t) = 2t - 1$ we already see that the line is contained in the plane $x = 1$. To get another plane, note that $y = t = (z + 1)/2$, and hence $y = (z + 1)/2$. Here is a picture of the points P, Q and the planes $x = 1$ and $2y - z = 1$.



Remark: We can get infinitely many planes containing the line by combining the equations $x - 1 = 0$ and $2y - z - 1 = 0$ with any coefficients a and b to get

$$a(x - 1) + b(2y - z - 1) = 0.$$

For example, taking $a = 1$ and $b = -1$ gives $x - 2y + z = 0$. We observe that this plane does indeed contain the points P and Q .

Problem 2. Suppose that a parametrized path $\mathbf{r}(t) = (x(t), y(t), z(t))$ satisfies

$$\mathbf{r}(0) = (0, 0, 0),$$

$$\mathbf{r}'(0) = (0, 1, 2),$$

$$\mathbf{r}''(t) = (\cos t, -\sin t, -4\sin(2t)).$$

Use this information to compute $\mathbf{r}(t)$ for all t .

First we integrate $\mathbf{r}''(t)$ to get $\mathbf{r}'(t)$:

$$\begin{aligned}\mathbf{r}'(t) &= \int \mathbf{r}''(t) dt \\ &= \left(\int \cos t dt, \int -\sin t dt, \int -4\sin(2t) dt \right) \\ &= (\sin t + c_1, \cos t + c_2, 2\cos(2t) + c_3).\end{aligned}$$

Substituting $t = 0$ gives $(0, 1, 2) = \mathbf{r}'(0) = (c_1, 1+c_2, 2+c_3)$. This implies that $c_1 = c_2 = c_3 = 0$ and hence $\mathbf{r}'(t) = (\sin t, \cos t, 2\cos(2t))$. Then integrate $\mathbf{r}'(t)$ to get $\mathbf{r}(t)$:

$$\begin{aligned}\mathbf{r}(t) &= \int \mathbf{r}'(t) dt \\ &= \left(\int \sin t dt, \int \cos t dt, \int 2\cos(2t) dt \right) \\ &= (-\cos t + c_4, \sin t + c_5, \sin(2t) + c_6).\end{aligned}$$

Substituting $t = 0$ gives $(0, 0, 0) = \mathbf{r}(0) = (-1 + c_4, c_5, c_6)$, hence $c_4 = 1$ and $c_5 = c_6 = 0$. We conclude that

$$\mathbf{r}(t) = (1 - \cos t, \sin t, \sin(2t)).$$

It turns out that this curve returns to its starting point at $t = 2\pi$, then it cycles around. Here is a picture:

